



MICROECONOMICS

COST

Chapter 6 | Lecture 3

Cost

Explicit

Implicit X

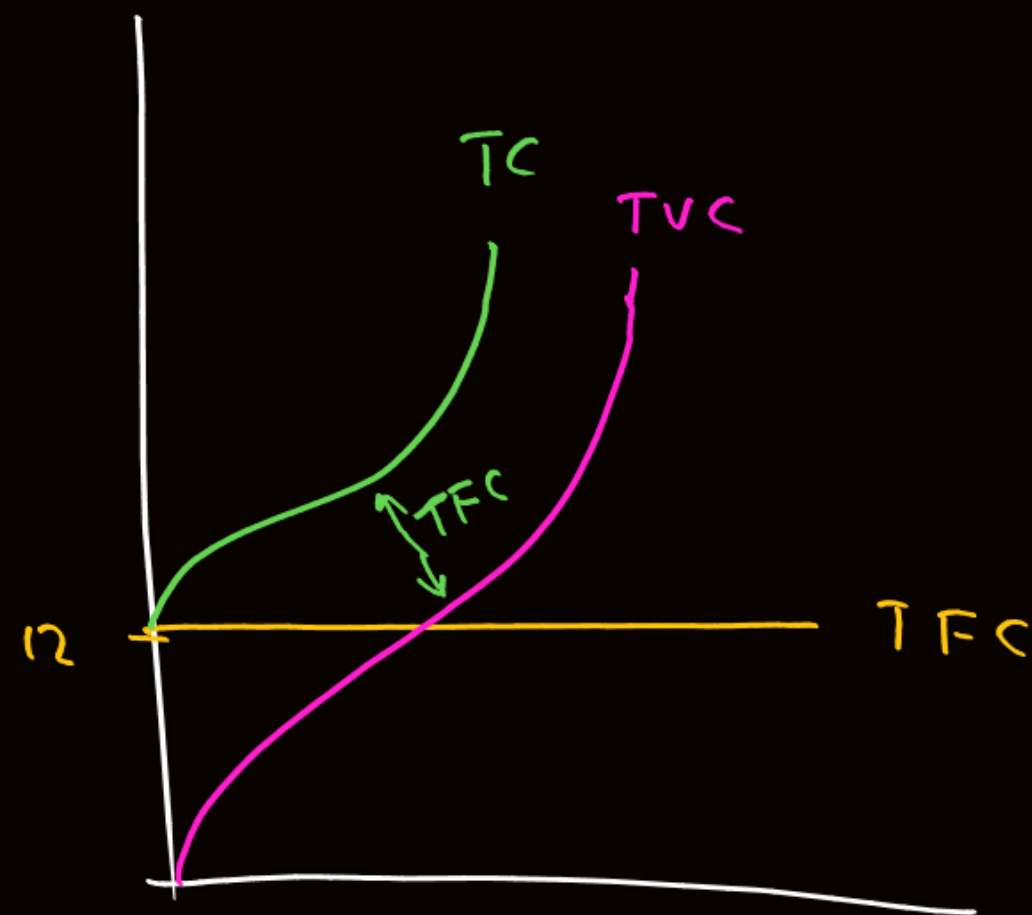
$$C = f(a)$$

Ex: Rent Interest capital

	TFC
0	12
1	12
2	12
3	12
4	12

	TVC
0	0
1	6
2	10
3	15
4	25

	TC
0	12
1	18
2	22
3	27
4	37



$$\frac{10}{Q^2} \text{ TFC} \downarrow \text{AFC} \quad 5$$

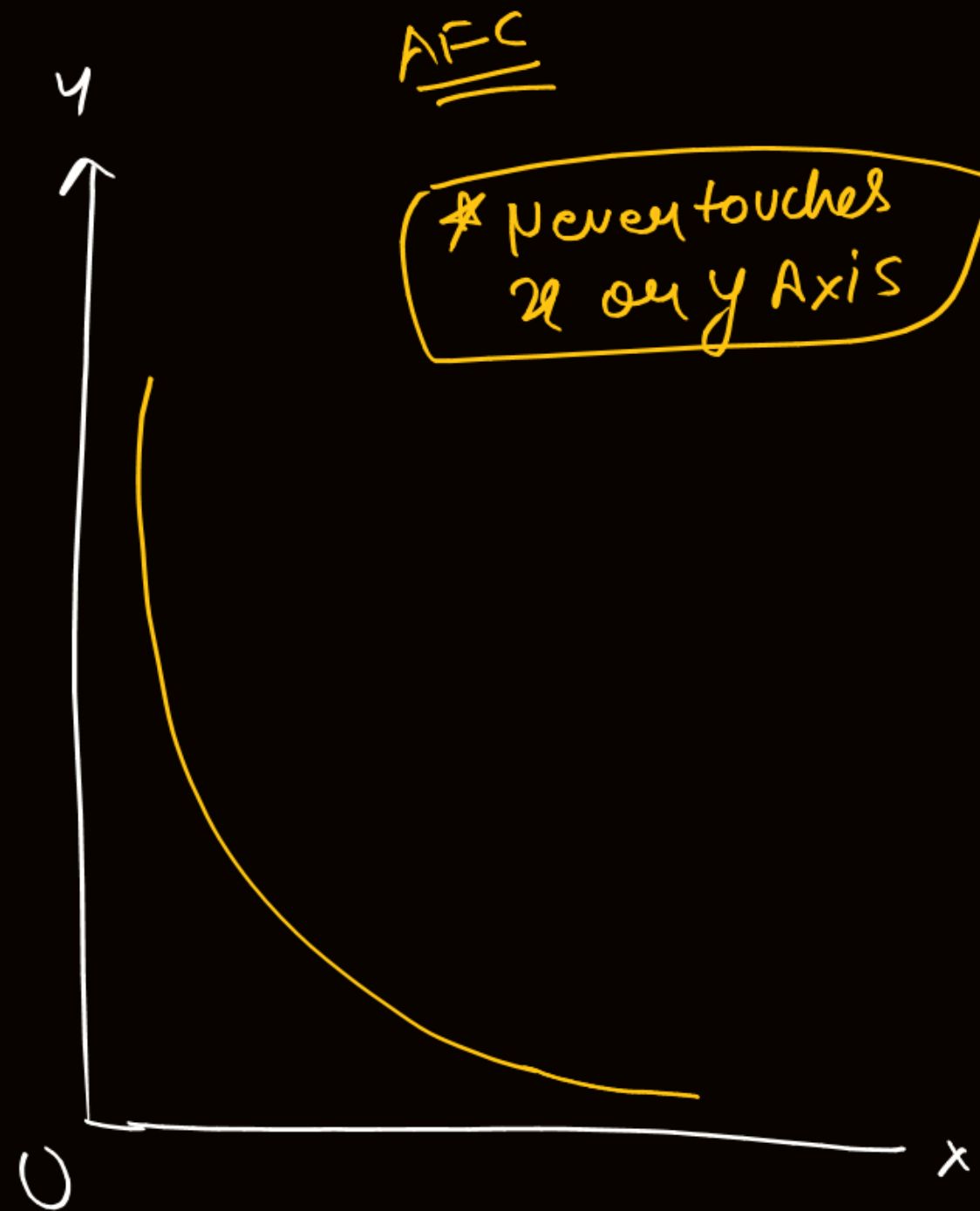
$$\frac{20}{2Q} \text{ TVC} \downarrow \text{AVC} \quad 10$$

$$= \frac{30}{Q^2} \text{ TC} \downarrow \text{AC} \quad 15$$

Q	TFC	AFC
0	12	$\frac{12}{0} = \infty$
1	12	$\frac{12}{1} = 12$
2	12	$\frac{12}{2} = 6$
3	12	$\frac{12}{3} = 4$
4	12	$\frac{12}{4} = 3$
5	12	$\frac{12}{5} = 2.4$
6	12	$\frac{12}{6} = 2$

Q_x

$Q \uparrow$ AFC $\downarrow$
Inverse Relation



$$TC = TFC + TVC$$

AVERAGE COST

➔ Average Fixed Cost (AFC)

Average fixed cost refers to the per unit fixed cost of production.

$$AFC = TFC \div Q$$

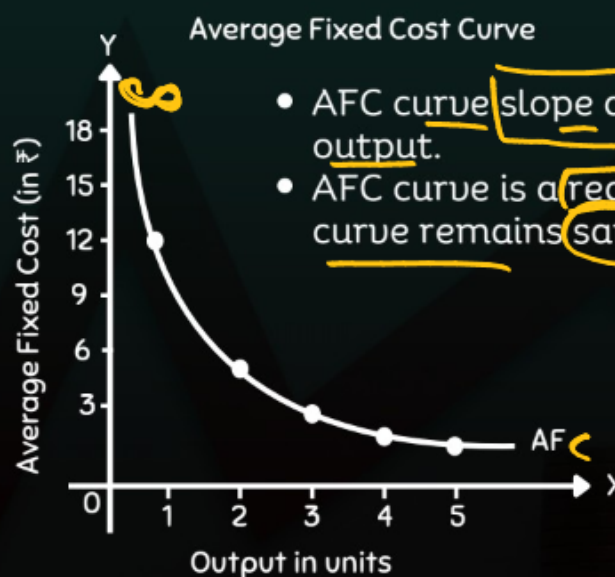
Average Fixed Cost

Output (in units)	TFC (₹)	AFC (₹) $TFC \div \text{Output} = AFC$
0	12	$12 \div 0 = \infty$
1	12	$12 \div 1 = 12$
2	12	$12 \div 2 = 6$
3	12	$12 \div 3 = 4$
4	12	$12 \div 4 = 3$
5	12	$12 \div 5 = 2.40$

6

12

2



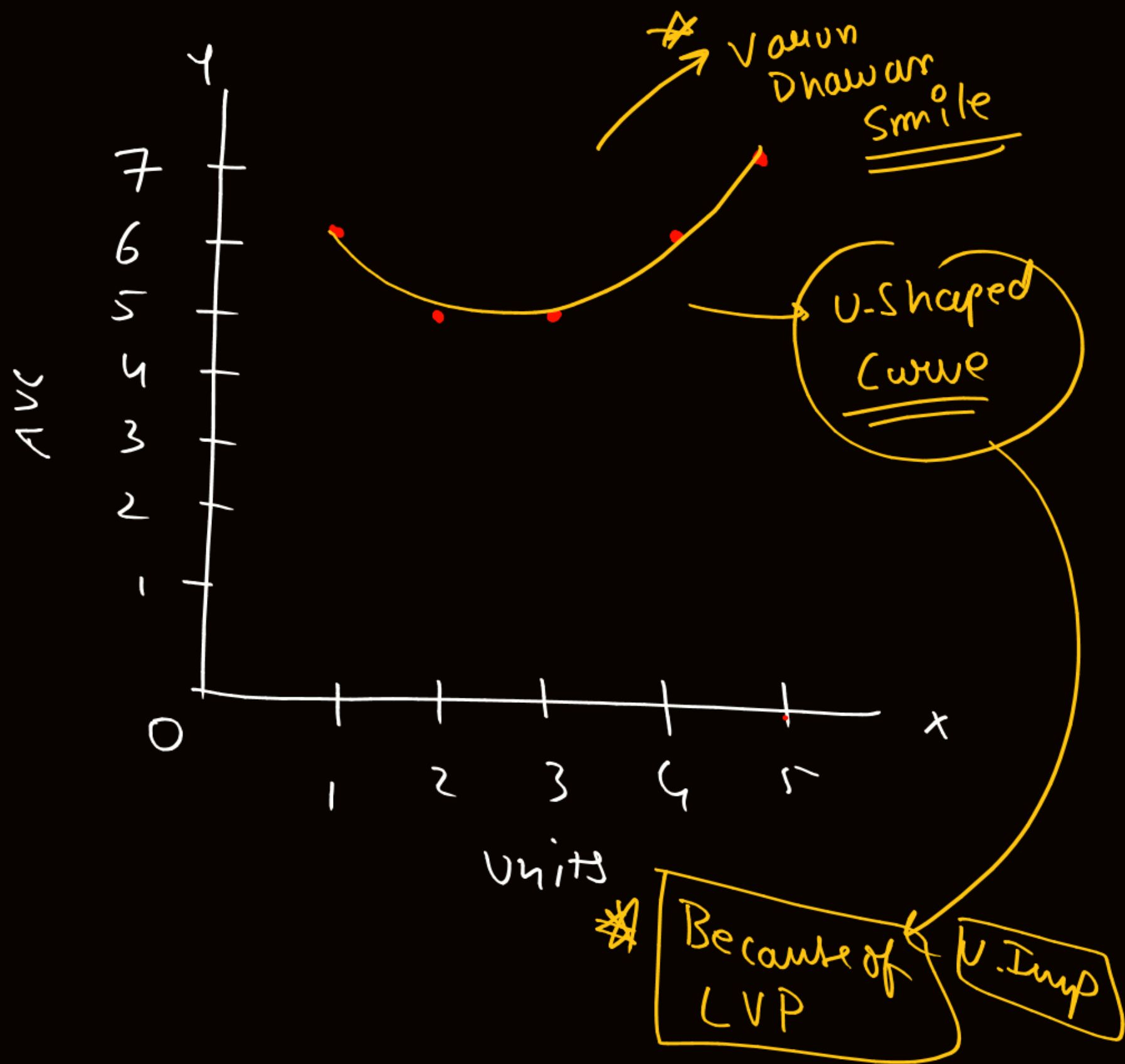
- AFC curve slope downwards as AFC falls with increase in output.
- AFC curve is a rectangular hyperbola, i.e. area under the curve remains same at all points.

↗ AFC ↘



- AFC does not touch any of the axes
 - AFC can never touch the X-axis as TFC can never be zero.
 - AFC curve can never touch the Y-axis because at zero level of output, TFC is a positive value and any positive value divided by zero will be an infinite value.

Q	TVC	AVC
0	0	
1	6	$\frac{6}{1} = 6$
2	10	$\frac{10}{2} = 5$
3	15	$\frac{15}{3} = 5$
4	24	$\frac{24}{4} = 6$
5	35	$\frac{35}{5} = 7$



➤ Average Variable Cost (AVC)

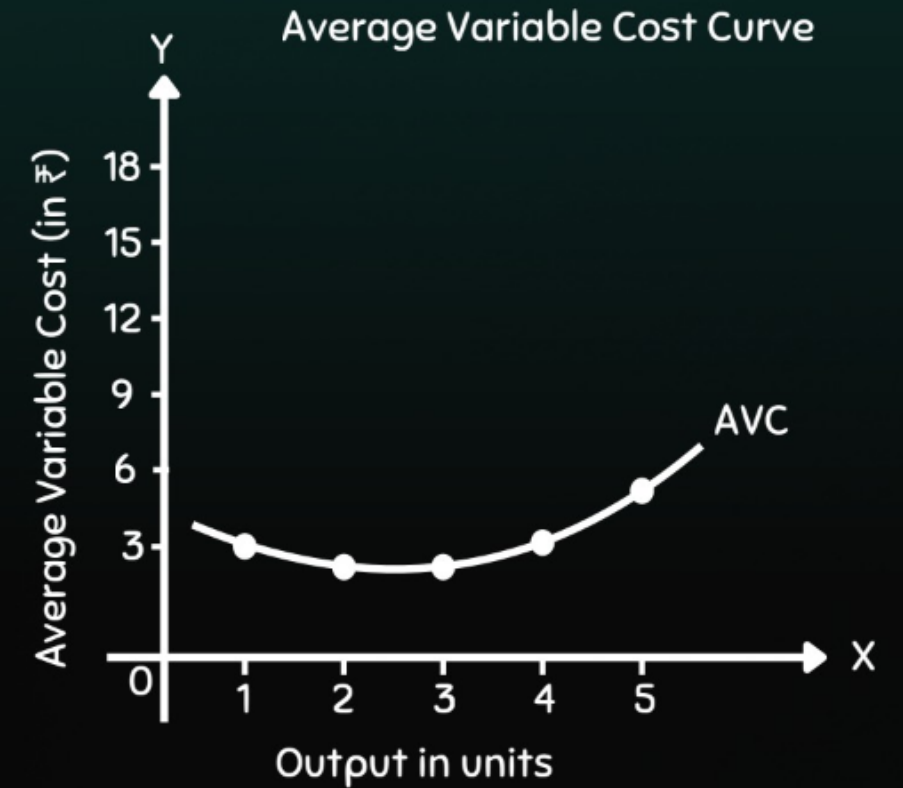
Average variable cost refers to the per unit variable cost of production.

$$AVC = TVC \div Q$$

Average Variable Cost

Output (in units)	TVC (₹)	AVC (₹) $TVC \div \text{Output} = AVC$
0	0	-
1	6	$6 \div 1 = 6$
2	10	$10 \div 2 = 5$
3	15	$15 \div 3 = 5$
4	24	$12 \div 4 = 6$
5	35	$35 \div 5 = 7$

AVC curve is a U-Shaped due to the application of Law of Variable Proportions.

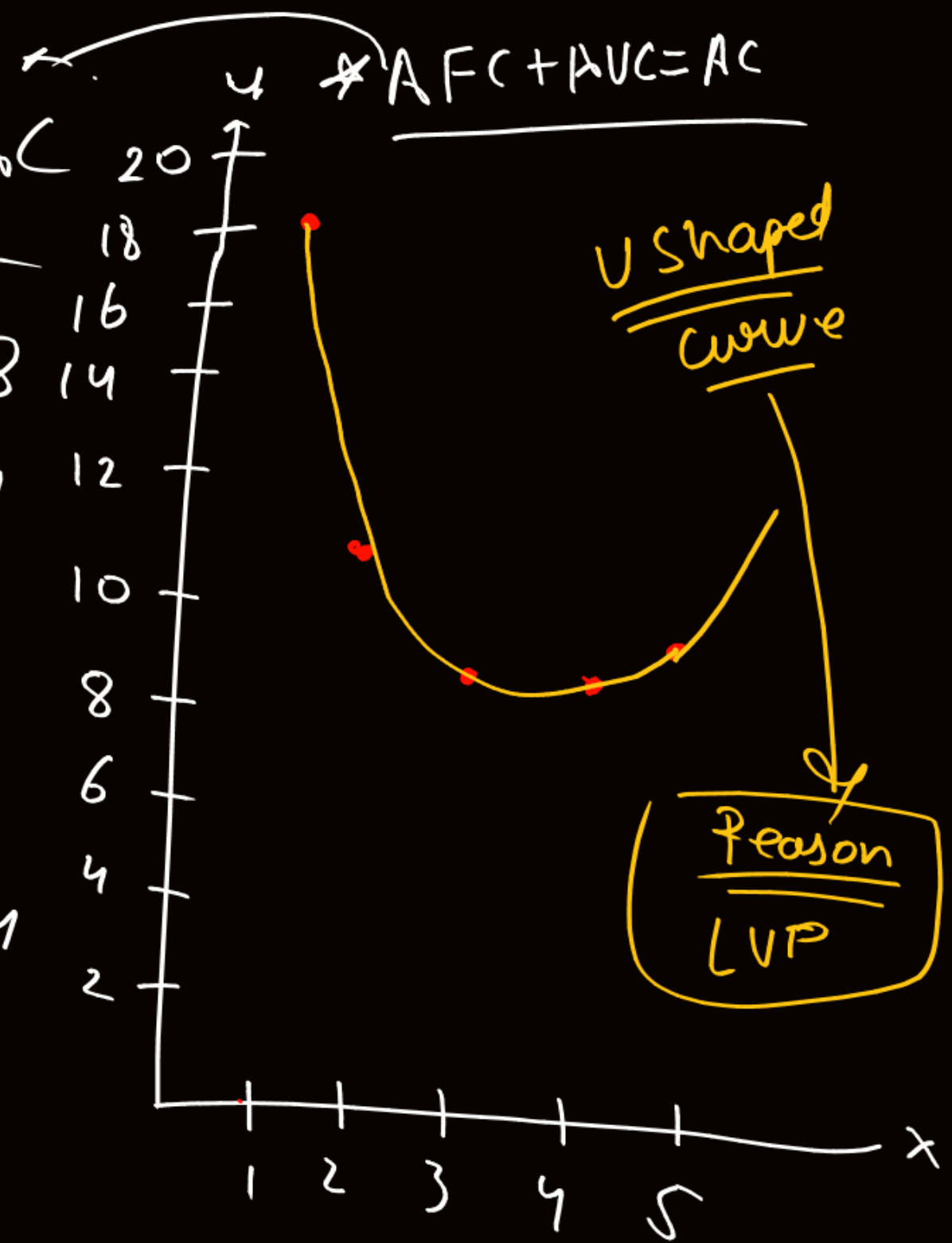


①

Q	TC	AC
0	12	-
1	18	18
2	22	11
3	27	9
4	36	9
5	47	9.4

②

AVC	AFC	AC
-	∞	-
6	12	18
5	6	11
5	4	9
6	3	9
7	2.4	9.4



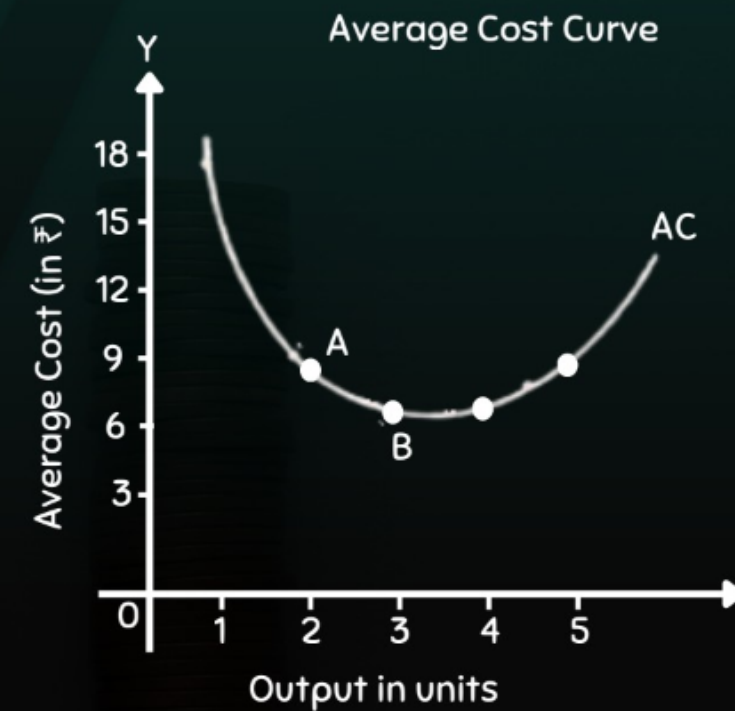
➤ Average Total Cost (ATC) or Average Cost (AC)

Average cost refers to the per unit total cost of production.

$$AC = TC \div Q$$

Average Cost

Output (in units)	AFC (₹)	AVC (₹)	AC (₹) TVC + Output = AVC
0	∞	-	-
1	12	6	$12 + 6 = 18$
2	6	5	$6 + 5 = 11$
3	4	5	$4 + 5 = 9$
4	3	6	$3 + 6 = 9$
5	2.40	7	$2.40 + 7 = 9.40$



U shaped
curve

Average cost is also defined as the sum of average fixed cost (AFC) and average variable cost (AVC), i.e., $AC = AFC + AVC$.

AFC

12

6

4

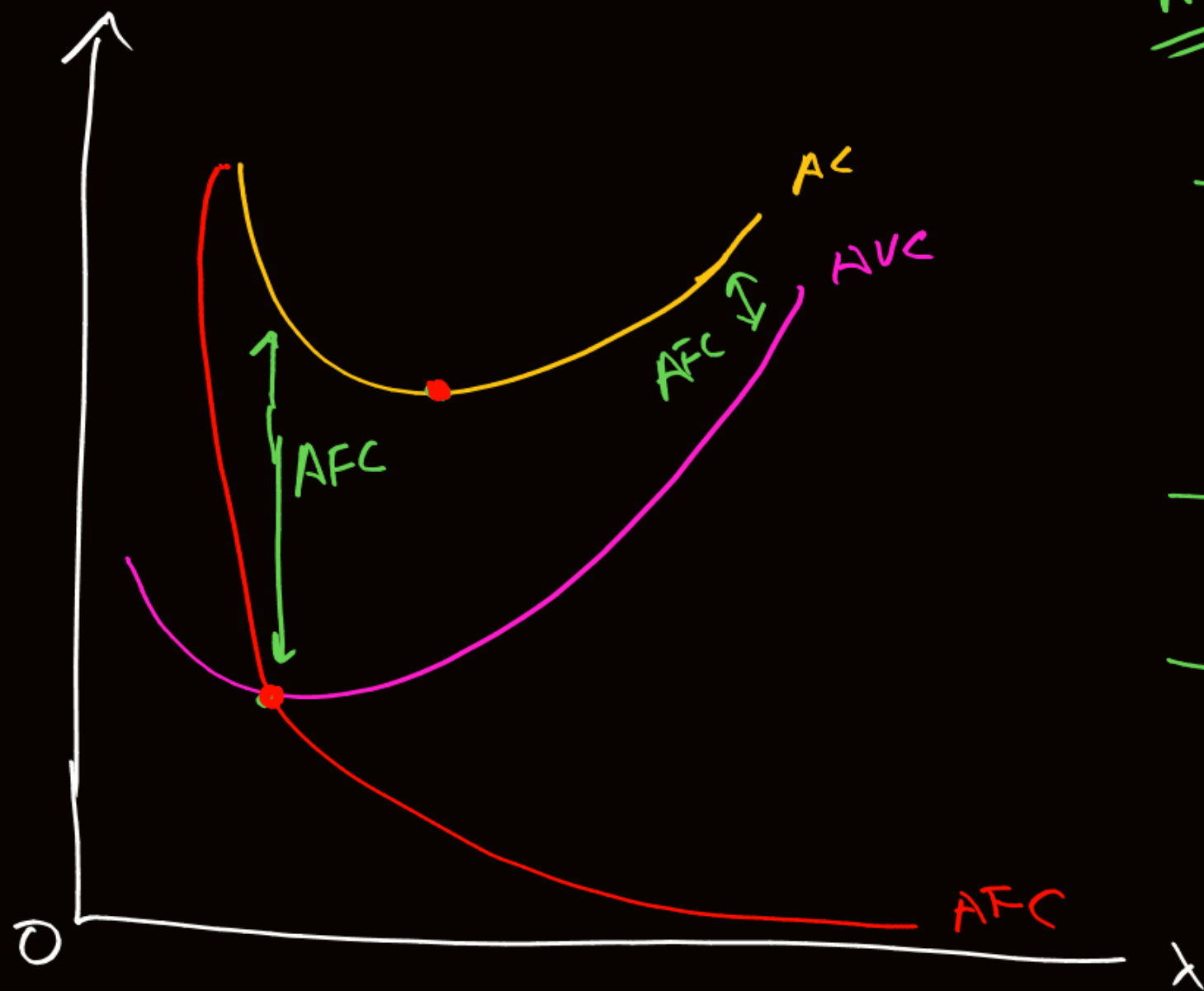
3

2.4

2

0.24

0.02



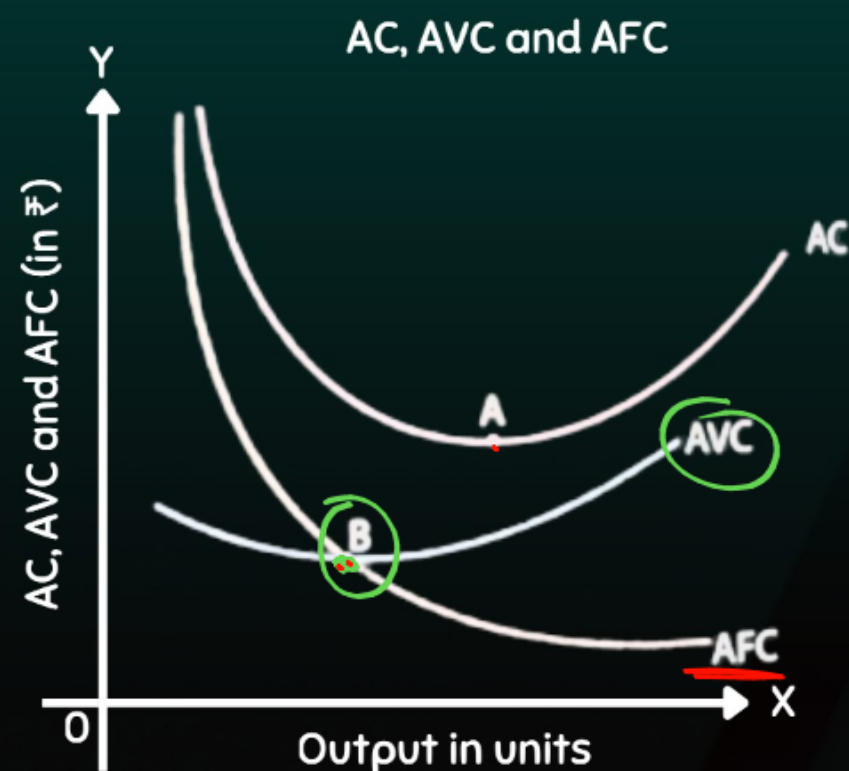
AC, AVC, AFC Relationship

→ AC is always Above AVC & AFC because AC is Sum of AFC & AVC.

→ AFC intersect AVC at it's Minimum.

→ Difference between AC & AVC is AFC and AC & AVC will never intersect each other because AFC can never be 0.

Average Total Cost (ATC) or Average Cost (AC)



- AC curve will always lie above the AVC curve (See Fig. 6.7) because AC, at all levels of output includes both AVC and AFC.
- AVC reaches its minimum point (point 'B') at a level of output lower than that of AC (point 'A') because when AVC is at its minimum point, AC is still falling because of falling AFC.
- As the output increases, the gap between AC and AVC curves decreases, but, they never intersect each other. It happens because the vertical distance between them is AFC, which can never be zero.

Class 11 ECONOMICS — Chapter 5: PRODUCTION FUNCTION

Published

10 Questions

40 Marks

20 Min

STUDENT	TOTAL SCORE	MEGA TEST	Correct	Incorrect	Skipped	Time
TAMANNA	80.00/80	80/80	20	0	0	7m 15s
DISHA MONDAL	80.00/80	80/80	20	0	0	8m 11s
KARTIK SINGH	80.00/80	80/80	20	0	0	7m 56s
ANERI	80.00/80	80/80	20	0	0	10m 29s
ANKIT THE LEGEND	80.00/80	80/80	20	0	0	6m 10s
PURVA SINGH	80.00/80	80/80	20	0	0	17m 21s
TEJAS PAL	80.00/80	80/80	20	0	0	7m 4s
SHRISTI	76.00/80	76/80	19	1	0	17m 23s
BANDANA	76.00/80	76/80	19	1	0	22m 44s
AARATI	76.00/80	76/80	19	1	0	6m 46s

Class 11 ECONOMICS — Chapter 5: PRODUCTION FUNCTION

Published

10 Questions

40 Marks

20 Min

STUDENT	TOTAL SCORE	MEGA TEST	Correct	Incorrect	Skipped	Time
ANSHIKA SHARMA	76.00/80	76/80	19	1	0	5m 32s
DHANYA GARG	76.00/80	76/80	19	1	0	6m 12s
HEENA	76.00/80	76/80	19	1	0	5m 6s
VISHWADEEPTI MISHRA	76.00/80	76/80	19	1	0	8m 1s
BHUMIKA TOLANI	76.00/80	76/80	19	1	0	4m 3s
SAHIBA AFROZ	76.00/80	76/80	19	1	0	5m 25s
KHUSHI GAUR	76.00/80	76/80	19	1	0	7m 17s
VANSHIKA KAUSHIK	76.00/80	76/80	19	1	0	59s
AKANSHA SATWANI	76.00/80	76/80	19	1	0	16m 46s
ANUUUU	76.00/80	76/80	19	1	0	3m 23s

**THANK
YOU**

Stay Connected

KEEP LEARNING